

15.1: Numerical Measures:

A numerical measure is a parameter / statistic that allows us to describe it's central tendency or it's spread.

⇒ Central Tendency Measures: Mode, Median, Mean

⇒ Measures of spread: Range, Interquartile Range, Standard Deviation

	Adv	Disadv
Mean	Equally influenced. Appropriate for large / symmetrical samples	Affected by extreme values
Stnd. Devn.	Influenced Equally, large sample statistical modelling	Complicated w/o tech. More difficult than Stats
Median	Not all values needed Less affected by skew.	Hard to find in large data set
IQR	Not influenced by Outliers / skew Easy for small samples.	Difficult with many values (no tech)
Mode.	Easy to find	Multiple modes may exist
Range	Easy to calculate	Influenced by extremes

Median & IQR:

⇒ The Median is the middle observation of data

⇒ The IQR is the range of the middle 50% of the data

This is reliable as it reduces effect of extreme values.

⇒ The Median is the $\frac{1}{2}(n+1)^{th}$ Observation.

⇒ The IQR is the difference in Q1 ($\frac{1}{4}(n+1)^{th}$ observation) & Q3 ($\frac{3}{4}(n+1)^{th}$ Observation)

To find this, linear interpolation of a cumulative frequency table can be used. This can also be applied on a graph

The Mean & Standard Deviation

Note: $n = \sum f$, $\sum x = \sum xf$, $\sum x^2 = \sum x^2 f$. Use midpoints for x vals.

Mean: Measure of central tendency that can be affected by extreme values (as all values are included).

For a population: $\mu = \frac{\sum x}{\sum f}$

For a sample: $\bar{x} = \frac{\sum x}{\sum f}$

Standard Deviation: Measure of spread around the mean. SD^2 = variance

For a population:

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

or

$$\sigma = \sqrt{\frac{\sum (x-\mu)^2}{n}}$$

For a sample:

$$s = \sqrt{\frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)}$$

or

$$s = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}$$

Outliers:

A value can be said as an outlier if it doesn't lie in the ranges:

$$\left(Q1 - \tfrac{3}{2}(Q_3 - Q_1),\ Q3 + \tfrac{3}{2}(Q_3 - Q_1)\right)$$

$$\left(\frac{\sum x}{\sum f} - 2s,\ \frac{\sum x}{\sum f} + 2s\right)$$

They should not be omitted, unless told otherwise

Comparing Distributions:

Should compare measures of central tendency & spread, to state respective components.

15.2: Data Representation

Data may be qualitative (categorical) or quantitative (numerical)
If data is quantitative, it can be discrete (countable) or continuous (measurable)

Qualitative Data Representation:
Pictograms: ○ ○ △ (○ = 4 Pizzas)
Dot plots:
Bar Charts (Bars don't touch!)
Pie Charts:

Continuous, Qualitative Data Representation:
Grouped Freq. Tables
Cumulative Freq. Diagrams
Box plots.
Histograms

Discrete, Quantitative Data Representation:
Bar Charts
Line Charts
Frequency Tables
Stem & Leaf Diagrams
Box-and-whisker plots.

1	2	4	6
2	4	5	8
3	1	2	3
4	8	9	9
5	1	1	5

1|2 shows 12

Cumulative Frequency graphs:
Plotted coordinates are upper value of interval (x), and frequency (y)
All points will be connected back to (1st Bottom Interval, 0)
Data must have true boundaries

Time Series:
Model trends in data over time. Good for quantitative data over discrete time points.

Misinterpretation:
Identify ambiguous / misleading scales, appreciate how a diagram can misrepresent data.
Read wording carefully, check scale. 1% ≠ 1#. Think Contextually

15.3: Bivariate Data:

Bivariate data has 2 variables.
Use a scatter graph

Correlation
Strength of linearity in data, and the direction of the relationship. Agreement on linearity.
It can be described as strong, weak, positive, negative, or none.

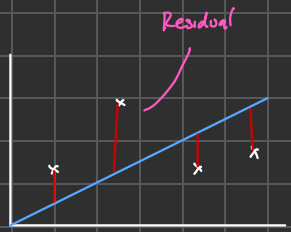
Product Moment Correlation Coefficient:
PMCC is a measure of the strength of correlation in bivariate data.

$r = 1 \Rightarrow$ Perfect, +ve, strong correlation.
$r = -1 \Rightarrow$ Perfect, -ve, strong correlation.
$r = 0 \Rightarrow$ No correlation.

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sqrt{\left[\sum x^2 - \frac{(\sum x)^2}{n}\right]\left[\sum y^2 - \frac{(\sum y)^2}{n}\right]}}$$

Spearman's Rank Correlation Coefficient:
The PMCC of ranked data & is a measure of strength of association between data

Regression:
Use of variables, independent & dependent, to form a line of best fit.
Independent - Variable being controlled / changed
Dependent - The variable that is being recorded



Regression lines:
If PMCC is close to ± 1, an equation of line of best fit can be found.

For a regression line:
$\sum(\text{residuals}) = 0$, $\sum(\text{residuals})^2$ is to be minimised
A regression line must include point $(\bar{x}, \bar{y})$
Residuals can be compared to find outliers. DO NOT IGNORE OUTLIERS!
Use mode $y = a + bx$ on calculator. This can be used to predict values

Predictions
Interpolation: Predicted x-value is within range of pre-observed x values. If PMCC is close to ± 1, the prediction is good
Extrapolation: Predicted x-value is outside range of pre-observed x values. Generally, this is unreliable.